SEEMANDHAR JAIN

+1 (408) 690-8037 | sejain@ucsd.edu | Google Scholar | LinkedIn | seemandharjain.com | GitHub

EDUCATION

- University of California San Diego — CA, USA
  PhD in Computer Science; GPA: 4/4 — July 2024 - Present
  Thesis Advisor: Dr. Manmohan Chandraker
  Thesis Title: Agentic Scientific Discovery for Physical AI.

- University of Illinois Urbana-Champaign — IL, USA
  Master of Science in Computer Science; GPA: 3.92/4 — Aug 2022 - May 2024
  Thesis Advisor: Dr. David Forsyth
  Thesis Title: Single-shot 3d Reconstruction and Convex Decomposition of Indoor Scenes
  Siebel Scholar, Class of 2024

- IIT Indore — MP, India
  Bachelor of Technology in Computer Science and Engineering; GPA: 3.6/4 — July 2017 - May 2021
  Minor in Humanities and Social Sciences; GPA: 3.6/4

SELECTED PUBLICATIONS

- **S. Jain**, M. Chandraker. RoboMetaHarness*: Automated Harness Discovery for Frozen Robot Policies.* Submitted to **CVPR 2027**.
- **S. Jain**, K.G, K.G, M. Chandraker. Nerfify*: Multi-Agent Framework for Turning NeRF Papers into Code.* **CVPR 2026, Highlight**.
- **S. Jain**, M. Chandraker. *Who Wrote This Paper? Autonomous Scientific Discovery for 3DGS Research.* **NeurIPS 2026**.
- **S. Jain**, K. Gupta, M. Chandraker. SPLATIFY*: Reproduce, Discover, Innovate! From Papers and Ideas to Trainable 3DGS Code.* Submitted to **CVPR 2027**.
- K. Gupta, G. Joshi, Y.R. Chen, **S. Jain**, H. Li, I. Mehta, M. Chandraker. *Reflecting Tacit Video Expertise in Procedural Material Generation.* **ECCV 2026**.
- V. Vavilala, **S. Jain**, R. Vasanth, D. Forsyth, A. Bhattad. *Generative Blocks World: Moving Things Around in Pictures.* **ICCV 2025 Workshop on Wild 3D**.
- Xiao Zhang, **S. Jain**, Michael Maire, A. Bhattad, D. Forsyth. *Latent Intrinsics Emerge from Relighting Training.* NeurIPS, 2024. [link]
- V. Vavilala, **S. Jain**, R. Vasanth, A. Bhattad, D. Forsyth. *Blocks2World: Controlling Realistic Scenes with Editable Primitives.* [link]
- **S. Jain**, P. Jain, PK. Upadhyay, JM. Moualeu, A. Srivastava. *An Energy Efficient Fault-Anomaly Detection in Wireless Body Area Networks.* ACM Transactions on Computing for Healthcare, 2022. [link]
- P. Jain, **S. Jain**, OR. Zaïane, A. Srivastava. *Anomaly Detection in Resource Constrained Environments with Streaming Data.* IEEE Transactions on Emerging Topics in Computational Intelligence, 2021. [link]
- **S. Jain**, AA. Shastri, K. Ahuja, Y. Busnel, NP. Singh. *Cube Sampled K-Prototype Clustering for Featured Data.* IEEE 18th India Council International Conference, 2021. [link]

WORK/RESEARCH EXPERIENCE

- **University of California San Diego**, CA, USA — July 2024 – Present
  Graduate Research Assistant under Dr. Manmohan Chandraker
  - Proposed SplatLab (NeurIPS 2026), a multi-agent framework for end-to-end scientific discovery, integrating literature review, hypothesis generation, method implementation, experimental validation, and manuscript synthesis.
  - Developing RoboMetaHarness, a training-free framework that automatically discovers executable harnesses for frozen robot policies through failure-guided experiments, policy-interface conditioning, and interaction-aware composition to improve generalization to unseen tasks.
  - Researching vision-language-action models (VLAs) and world-action models (WAMs) for embodied agents, focusing on long-horizon reasoning, planning, and robotic manipulation.
  - Proposed Nerfify (CVPR 2026 Highlight), a multi-agent system that translates NeRF research papers into executable implementations through coordinated understanding, coding, and verification.
  - Proposed Splatify, extending paper-to-code automation to 3D Gaussian Splatting through a knowledge-discovery pipeline connecting research understanding, implementation, and experimentation.

- **Adobe**, San Jose, CA, USA — June 2026 - Sept. 2026
  Research Scientist Intern (Mentors: Yannick Hold-Geoffroy, Manmohan Chandraker)
  - Developing an autonomous multi-agent pipeline that procedurally composes *mashup* 3D scenes in Unreal Engine 5.5 and auto-curates large-scale synthetic datasets with dense, pixel-accurate ground truth.
  - Improving downstream world models by auto-curating targeted training data, closing the loop between scene synthesis and model performance.

- **Google**, New York, NY
  Student Researcher, BigML Team — Jun 2025 - Sept 2025
  - Optimized SANA diffusion models for the image/video generation pipeline, achieving **1.3× speedup** in sampling/inference while maintaining visual fidelity.
  - Developed a Gemma 2-based image captioning pipeline that improved LLM-conditioned generation quality, reducing FID by **16%**.
  - Enabled faster frame synthesis for large-scale video generation by streamlining the diffusion sampling process.

- **Aireal**
  Research Scientist, Computer Vision May 2024 - Sep 2024
  - Played a key role in raising **$5 million** in Series A funding by developing a novel single-image-to-3D texture mesh technique.
  - Developed an AI-powered interior design tool called Livvy that helps users transform inspiration images into their existing space.
  - Conducted research on the 3D reconstruction of rooms using diffusion models and NeRFs for virtual furniture removal/addition.
- **University of Illinois Urbana-Champaign**, IL, USA, Aug 2022 - May 2024
  Graduate Teaching/Research Assistant under Dr. David Forsyth
  - Advanced 3D scene reconstruction, proposed *Blocks2World, 3DControlNet* using deep learning, and developed scenes segmentation.
  - Integrated generative models like stable diffusion and 3D-aware image editing applications, using mesh and point cloud.
  - Generated outdoor videos using diffusion models trained on our self-image-captioned indoor image dataset with multimodal LLMs.
  - Conducted intrinsic image decomposition into its albedo and shading components and performed relighting using diffusion models.

  Teaching Assistant for CS 225 (Data Structure and Algorithms), CS 128 (Intro to CS II), and CS 441 (Applied Machine Learning).
- **DOCOMO Innovations, Inc.** California, USA
  Research Intern May 2023 - August 2023
  - Developed a GPT2-based realtabformer model to fill missing data in datasets, which improved model prediction.
  - Designed and implemented a practical stereo depth system for videos, enhancing 3D depth perception.
  - Generated ads using Stable Diffusion, employing mask-based inpainting with LaMa and fine-tuned diffusion models on specific data.
- **HealthAIgnite**
  Founder Jan 2022 - Present
  - Founded and currently manages HealthAIgnite, a leading healthcare AI consulting company focusing computer vision technologies.
  - Established partnerships with key healthcare institutions and MedTech companies, accelerating the adoption of AI solutions.
- **Salesforce** Hyderabad, India
  AMTS Software Developer (Artificial Intelligence) July 2021 - August 2022
  - Worked as a software developer in the CnP team, focusing on the development of Salesforce Products and Einstein AI.
  - Implemented tax schema for EMEA region and performed full-stack development for feature enhancement on the Force.com platform.
  - Automated core patch regression processes and deployed APIs using Mulesoft and Heroku, resulting in significant time saving.
  - Served as a Scrum Master, leading the major release of the FnE product that added numerous new features for customers.

  Software Developer Intern May 2020 - July 2020
  - Automated the manual process of estimating the time required to complete a Mulesoft project by implementing a machine learning algorithm on Heroku, creating REST API using Django, and developing front-end on the lightning platform using Aura and Apex.
- **University of Witwatersrand and University of Alberta**, South Africa and Canada
  Research Assistant under Dr. Osmar Zaiane, Dr. Jules M. Moualeu and Dr. Abhishek Srivastava, Jan 2019 - May 2021
  - Conducted research on an energy-efficient health monitoring approach with wireless body area networks (WBAN). Proposed an ML innovative approach to energy conservation and detection of alarming health status.
  - Designed an automated surveillance system using ML models that utilize wireless sensor networks for detecting forest fires.
  - Implemented a two-tier anomaly detection algorithm to identify and predict adverse conditions from health parameter readings.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, SQL, MATLAB, R, Apex.
Generative AI & Agents: LLMs, Multi-Agent Systems, Automated Scientific Discovery, Diffusion Models, Image/Video Generation.
Computer Vision & Embodied AI: 3D Reconstruction, NeRFs, 3D Gaussian Splatting, World Models, Embodied Policy Learning.
ML Frameworks & Computing: PyTorch, JAX, TensorFlow, CUDA, Docker, Kubernetes, Git, AWS (EC2, Lambda, S3), Azure.
Web & Databases: Django, Flask, React, PostgreSQL, MongoDB.

ACHIEVEMENTS

- Awarded **Siebel Scholar**, Class of 2024 by Siebel foundation from UIUC.
- Interviewed by **Times of India** news for research on wildfire detection algorithm (2022).
- **Best BTP award** from Computer Science Department among the graduating students (2021).
- Qualified for Onsite Round of ACM ICPC 2019 and received honorable mention in ICPC Amritapuri regionals.
- Secured **1st** position among the **top 50** finalists of **DRONA 2018**: A management competition by IIM INDORE (2018).
- Secured **3rd** rank in Intra IIT Tech Meet Robo Race (2017).
- Secured All India Rank **1st** in IIT JEE Advanced (2017) in Mathematics Section with a score 122/122.
- Secured **1400** rank in IIT JEE Advanced (2017) among a million students.
- Secured **AIR 43** in technothlon hosted by IIT GUWAHATI (2016).
- Secured **State rank 2** and **International rank 26** in International Mathematics Olympiad (2015).
- UCMAS (Mental Math and Abacus) degree scholar, and secured 9th rank in UCMAS global competition (2013).